Probabilistic AI

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1. Math / Probability

- **Bernoulli:** (biased) coin flip, $\mathbb{P}(X{=}H) = p$, $\mathbb{P}(X{=}T) = (1-p)$, yields *Binomial* distr. $\mathbb{P}(X{=}k) = \binom{n}{k} p^k (1-p)^{n-k}$
- **Categorical:** (biased) m-sided dice, $\mathbb{P}(X{=}i) = p_i, p_i \geq 0, \sum p_i = 1$, yields *Multinomial* distr.
- **Chain rule:** $\mathbb{P}(X \wedge Y) = \mathbb{P}(X \mid Y)\mathbb{P}(Y)$, $\mathbb{P}(X_1, \dots, X_n) \equiv \mathbb{P}(X_{1:n}) = \mathbb{P}(X_1)\mathbb{P}(X_2 \mid X_1) \dots \mathbb{P}(X_n \mid X_{1:n-1})$
- **Marginal:** $\mathbb{P}(X_i) = \sum_{x_1} \dots \sum_{x_n} \mathbb{P}(x_{1:i-1}, X_i, x_{i+1:n})$, $\mathbb{P}(X_{1:i-1}, X_{i+1:n}) = \sum_{x_i} \mathbb{P}(X_{1:i-1}, x_i, X_{i+1:n})$
- $X \perp Y$: $\mathbb{P}(X,Y) \overset{\text{chain}}{=} \mathbb{P}(X)\mathbb{P}(Y \mid X) \overset{\text{indep}}{=} \mathbb{P}(X)\mathbb{P}(Y)$
  $X \perp Y \mid Z$: $\mathbb{P}(x,y|z) = \mathbb{P}(x|z)\mathbb{P}(y|z)$, $\mathbb{P}(X|Y,Z) = \mathbb{P}(X|Z)$

Bayes: $\mathbb{P}(X \mid Y) = \frac{\mathbb{P}(X,Y)}{\mathbb{P}(Y)}$, prior $\mathbb{P}(X)$, likel. $\mathbb{P}(Y|X)$, post. $\mathbb{P}(X|Y)$, evid. $\mathbb{P}(Y) \neq 0$

- $\mathbb{V}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$
  $\mathbb{V}[X+Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2\,\mathbb{C}\text{ov}(X,Y)$
- $\mathbb{C}\text{ov}(X,Y) = \mathbb{E}[(X-\mu_X)(Y-\mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$
  $\mathbb{C}\text{ov}(aX, bY) = ab\,\mathbb{C}\text{ov}(X,Y)$

1.1. Multivariate Gaussian

$\Sigma = \{\sigma_{ij}\}$

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})\right)$$

- $\sigma_{ij} = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \mathbb{C}\text{ov}(X_i, X_j) \overset{X_i \perp X_j}{=} 0$
- $\sigma_i^2 = \mathbb{E}[(X_i - \mu_i)^2] = \mathbb{V}[X_i]$

Gaussians are closed under cond., margin., mult. & add.

- $\mathbf{Y} := \mathbf{M}\mathbf{X} + \mathbf{X}' \sim \mathcal{N}(\mathbf{M}\boldsymbol{\mu} + \boldsymbol{\mu}', \mathbf{M}\Sigma\mathbf{M}^\top + \Sigma')$, $\mathbf{M} \in \mathbb{R}^{m \times d}$
- **Conditional:** $\mathbf{X}_A | X_B = \mathbf{x}_B \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}, \tilde{\Sigma})$ where $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu}_A + \Sigma_{AB}\Sigma_{BB}^{-1}(\mathbf{x}_B - \boldsymbol{\mu}_B)$ and $\tilde{\Sigma} = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$
  - RV X_A as model of X_B (1D): $X_A = aX_B + b + \varepsilon = \sigma_{AB}\sigma_B^{-2} \cdot X_B + (\mu_A - \sigma_{AB}\sigma_B^{-2}\mu_B) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_{A|B}^2)$
- **Complete the square:** If $p(\mathbf{x}) \propto \exp(-\frac{1}{2}\mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{x}^\top \mathbf{b})$, then $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \mathbf{A}^{-1}\mathbf{b}, \mathbf{A}^{-1})$

2. Bayesian Linear Regression — BLR

Ridge: $\varepsilon_{\text{RSS}}(\mathbf{w}, \lambda) = \|\mathbf{y} - \mathbf{X}^\top \mathbf{w}\|_2^2 + \lambda \mathbf{w}^\top \mathbf{w}$, $\lambda = \sigma_n^2 / \sigma_p^2$

Model: $y_i = \mathbf{w}^\top \mathbf{x}_i + \varepsilon = (\mathbf{X}\mathbf{w})_i + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$

Prior: $p(\mathbf{w}) = \mathcal{N}(0, \boldsymbol{\Lambda}^{-1}) \overset{\text{iid}}{\Longrightarrow} \mathcal{N}(0, \sigma_p^2 \mathbf{I}) \overset{\text{Ridge}}{\Longrightarrow} \mathcal{N}(0, \mathbf{I})$

Likelihood (iid): $p(y_i \mid \mathbf{X}, \mathbf{w}, \sigma_n) = \mathcal{N}(y_i; \mathbf{w}^\top \mathbf{x}_i, \sigma_n^2)$

Posterior: $p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \mathcal{N}(\mathbf{w}; \bar{\boldsymbol{\mu}}, \bar{\Sigma})$ with $(\hat{\mathbf{w}}_{\text{MAP}} = \dots)$

$\bar{\boldsymbol{\mu}} = (\mathbf{X}^\top \mathbf{X} + \sigma_n^2 \boldsymbol{\Lambda})^{-1} \mathbf{X}^\top \mathbf{y}$ and $\bar{\Sigma} = \sigma_n^2 (\mathbf{X}^\top \mathbf{X} + \sigma_n^2 \boldsymbol{\Lambda})^{-1}$

Predict: avg. acc. to *all* post. $\mathbf{w} \sim \mathcal{N}$, $p(y^* \mid \mathbf{X}, \mathbf{y}, \mathbf{x}^*) = \mathbb{E}_{\mathbf{w} \sim \text{post.}}[p(y^* \mid \mathbf{x}^*, \mathbf{w})] = \mathcal{N}\left(y^*; \bar{\boldsymbol{\mu}}^\top \mathbf{x}^*, \mathbf{x}^{*\top} \bar{\Sigma} \mathbf{x}^* + \sigma_n^2\right)$

Uncertainty: aleat. σ_n^2 (noise), epist. $\mathbf{x}^{*\top} \bar{\Sigma} \mathbf{x}^*$ (model)

Choose Hparams: (one option) Find λ via CV, estimate $\sigma_n^2 = \|\mathbf{y} - \mathbf{X}^\top \mathbf{w}\|_2^2$, then solve for $\sigma_p^2 = \frac{\sigma_n^2}{\lambda}$.

2.1. Recursive Bayesian updates — online BLR

$\mathbf{X}_{1:t}^\top \mathbf{X}_{1:t} = \|\mathbf{X}_{1:t-1}\|_2^2 + \|\mathbf{x}_{1:t}\|_2^2$ same for $\mathbf{X}^\top \mathbf{y}$ term.

Posterior is **recursive!** $p_{t+1}(\mathbf{w} \mid \boldsymbol{D}) = \frac{1}{Z} \cdot p(\mathbf{w}) \cdot p(\boldsymbol{D} \mid \mathbf{w}) = \frac{1}{Z} \cdot p(\mathbf{w} \mid (\mathbf{x}, y)_{1:t}) \cdot p((\mathbf{x}, y)_{t+1} \mid \mathbf{w})$

2.2. Kalman filters — KF

Location: $X_1, \dots, X_T$, Observ.: $Y_1, \dots, Y_T$, Prior: $\mathbb{P}(\mathbf{X}_1)$

- Motion model: $\mathbf{X}_{t+1} = \mathbf{F}\mathbf{X}_t + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \Sigma_x)$
- Sensor model: $\mathbf{Y}_t = \mathbf{H}\mathbf{X}_t + \eta_t$, $\eta_t \sim \mathcal{N}(0, \Sigma_y)$

Bayesian filtering in KFs: Start w/ $\mathbb{P}(\mathbf{X}_1)$; At time t:

1. **Conditioning:** $\mathbb{P}(\mathbf{X}_t \mid \mathbf{y}_{1:t}) \propto \mathbb{P}(\mathbf{X}_t \mid \mathbf{y}_{1:t-1}) \mathbb{P}(\mathbf{y}_t \mid \mathbf{X}_t)$
   Hint: Use formula for cond. Gaussians ($\mathcal{Y}_t = y_t$).
2. **Pred.:** $\mathbb{P}(\mathbf{X}_{t+1} \mid \mathbf{y}_{1:t}) = \int \mathbb{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) \mathbb{P}(\mathbf{x}_t \mid \mathbf{y}_{1:t}) \,\mathrm{d}\mathbf{x}_t$

3. Gaussian Processes — GP

Prediction: $f \sim GP(f; \mu, k)$, observe $y_i = f(\mathbf{x}_i) + \epsilon_i, \epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$.
Then $p(f \mid \mathbf{x}_{1:m}, y_{1:m}) = GP(f; \mu', k')$
where $\mu'(\mathbf{x}) = \mu(\mathbf{x}) + \mathbf{k}_{x,A}(\mathbf{K}_{AA} + \sigma_n^2 \mathbf{I})^{-1}(\mathbf{y}_A - \mu_A)$
and $k'(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') - \mathbf{k}_{x,A}(\mathbf{K}_{AA} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}_{x',A}^\top$

Forward sampling: Use chain rule on $p(f_1, \dots, f_n)$. Each sample is just a Gauss conditioned of prev. samples.

Marginal LL: instead of $p(\mathbf{y} \mid \mathbf{X}, \theta) \overset{\text{MAP}}{\approx} p(\mathbf{y} \mid \hat{f}, \theta)$ use
$\hat{\theta} = \arg\max_\theta p(\mathbf{y}_{\text{train}} \mid \mathbf{X}_{\text{train}}, \theta) = \int p(\mathbf{y} \mid \mathbf{X}, f, \theta)\, p(f \mid \theta) \,\mathrm{d}f$.

Fast GPs: GP prediction has cost $\mathcal{O}(n^3)$ ☹

- *Local* method: Use distance decaying kernel (e.g. RBF), only condition on points x': $|k(x,x')| \geq \tau$
- *Approximate* kernel $k(x,x') \approx \phi(x)^\top \phi(x') \to \mathcal{O}(nm^2 + m^3)$
- RFF: *Stationary* kernel $k(\mathbf{x}, \mathbf{x}') = \int_{\mathbb{R}^d} p(\omega) e^{\mathrm{j}\omega^\top(\mathbf{x}-\mathbf{x}')} \,\mathrm{d}\omega = \mathbb{E}_{\omega,b}[\mathbf{z}_{\omega,b}(\mathbf{x}) \cdot \mathbf{z}_{\omega,b}(\mathbf{x}')] \approx \frac{1}{m}\sum_i \dots = \mathbf{z}(\mathbf{x})^\top \mathbf{z}(\mathbf{x}')$ where $\omega \sim p(\omega)$, $b \sim U([0, 2\pi])$, $\mathbf{z}_{\omega,b}(\mathbf{z}) = \sqrt{2}\cos(\omega^\top \mathbf{x} + b)$
- Summarize data with values of f at *inducing points* $\mathbf{u}$
  $p(\mathbf{f}^*, \mathbf{f}) = \int p(\mathbf{f}^*, \mathbf{f}, \mathbf{u}) \,\mathrm{d}\mathbf{u} = \int p(\mathbf{f}^*, \mathbf{f} \mid \mathbf{u})\, p(\mathbf{u}) \,\mathrm{d}\mathbf{u}$
  $\approx q(\mathbf{f}^*, \mathbf{f}) = \int q(\mathbf{f}^* \mid \mathbf{u})\, q(\mathbf{f} \mid \mathbf{u})\, p(\mathbf{u}) \,\mathrm{d}\mathbf{u}$
  with $p(\mathbf{f}|\mathbf{u}) = \mathcal{N}(\mathbf{K}_{\mathbf{f}^*,\mathbf{u}}\mathbf{K}_{\mathbf{u},\mathbf{u}}^{-1}\mathbf{u}, \mathbf{K}_{\mathbf{f},\mathbf{f}} - \mathbf{Q}_{\mathbf{f},\mathbf{f}})$, $p(\mathbf{f}^*|\mathbf{u}) = \mathcal{N}(\mathbf{K}_{\mathbf{f}^*,\mathbf{u}}\mathbf{K}_{\mathbf{u},\mathbf{u}}\mathbf{u}, \mathbf{K}_{\mathbf{f}^*,\mathbf{f}^*} - \mathbf{Q}_{\mathbf{f}^*,\mathbf{f}^*})$ w/ $\mathbf{Q}_{\mathbf{a},\mathbf{b}} \equiv \mathbf{K}_{\mathbf{a},\mathbf{u}}\mathbf{K}_{\mathbf{u},\mathbf{u}}^{-1}\mathbf{K}_{\mathbf{u},\mathbf{b}}$

3.1. Kernels

scalar product: $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$

Valid kernel: **symmetr.** and **p.s.d.** ($\mathbf{x}^\top \mathbf{K}\mathbf{x} \geq 0\ \forall \mathbf{x}$ *or* pos. eigenvalues *or* pos. princip. minors). Must have a (pot. ∞-dim.) **feat. vec.** ϕ s.t. $k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x})^\top \phi(\mathbf{x}')$. *Stationary:* $k(\mathbf{x}, \mathbf{x}') = k(\mathbf{x} - \mathbf{x}')$, *isotropic:* $k(\mathbf{x}, \mathbf{x}') = k(\|\mathbf{x} - \mathbf{x}'\|_2)$.

Kernel construction: • $k_1 + k_2$ • $c \cdot k_1, c > 0$ • $k_1 \cdot k_2$ • $f(\mathbf{x}) k_1(\mathbf{x}, \mathbf{x}') f(\mathbf{x}')$ • $k(\phi(\mathbf{x}), \phi(\mathbf{x}'))$ with $\phi: \mathcal{X} \to \mathbb{R}^d$ • $g(k_1)$ with g: exp. *or* polyn. w/ all pos. coeff.

4. Variational Inference

Find approximate q s.t. $p(\theta \mid y) = \frac{1}{Z} p(\theta, y) \approx q(\theta \mid \lambda) \equiv q_\lambda$

4.1. Laplace Approximation (via 2nd Taylor)

$q(\theta) = \mathcal{N}(\theta; \hat{\theta}, \Lambda^{-1})$, $\hat{\theta} = \operatorname{amax}_\theta p(\theta \mid \mathbf{y})$, $\Lambda = -\nabla^2 \log p(\hat{\theta} \mid \mathbf{y})$

Predict: $p(y^* \mid \mathbf{x}^*, \mathbf{X}, \mathbf{y}) \approx \int p(y^* \mid f^*)\, q(f^*) \,\mathrm{d}f^*$ with $q(f^*) = \int p(f^* \mid \theta)\, q_\lambda(\theta) \,\mathrm{d}\theta$.

LA greedily fits mode, then matches curvature (over-conf.).

Appendix

BLR extended

Calculus

- $\int_a^b uv' \,\mathrm{d}x = uv|_a^b - \int_a^b u'v \,\mathrm{d}x$ • $\frac{\partial}{\partial x}\frac{g}{h} = \frac{g'h}{h^2} - \frac{gh'}{h^2}$
- $\frac{\partial}{\partial \mathbf{x}} f(\mathbf{x}) = \operatorname{diag}(f'(\mathbf{x}))$ • $\frac{\partial}{\partial \mathbf{x}}(\mathbf{b}^\top \mathbf{x}) = \frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{b}) = \mathbf{b}$
- $\frac{\partial}{\partial \mathbf{x}}(\mathbf{b}^\top \mathbf{A}\mathbf{x}) = \mathbf{A}^\top \mathbf{b}$ • $\frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{A}\mathbf{x}) = (\mathbf{A}^\top + \mathbf{A})\mathbf{x} \overset{*}{=} 2\mathbf{A}\mathbf{x}$ (*): $\mathbf{A}$ sym.
- $\frac{\partial}{\partial \mathbf{X}}(\mathbf{c}^\top \mathbf{X} \mathbf{b}) = \mathbf{c}\mathbf{b}^\top$ • $\frac{\partial}{\partial \mathbf{X}}(\mathbf{c}^\top \mathbf{X}^\top \mathbf{b}) = \mathbf{b}\mathbf{c}^\top$ • Tr. trick: $\mathbf{x}^\top \mathbf{A}\mathbf{x} \overset{*}{=} Tr(\mathbf{x}^\top \mathbf{A}\mathbf{x}) \overset{**}{=} Tr(\mathbf{x}\mathbf{x}^\top \mathbf{A}) = Tr(\mathbf{A}\mathbf{x}\mathbf{x}^\top)$
  (*): inner prod., (**): cyclic permut.
- $\frac{\partial}{\partial \mathbf{X}} Tr(\mathbf{X}^\top \mathbf{A}) = \mathbf{A}$ • $|\mathbf{X}^{-1}| = |\mathbf{X}|^{-1}$
- $\frac{\partial}{\partial \mathbf{X}} \log|\mathbf{X}| = \mathbf{X}^{-\top}$ • $\frac{\mathrm{d}}{\mathrm{d}x}|x| = \frac{x}{|x|} = \operatorname{sgn}(x)$
- $\frac{\partial}{\partial \mathbf{x}}\|\mathbf{x}\|_2 = \frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{x}) = 2\mathbf{x}$ • $\frac{\partial}{\partial \mathbf{x}}\|\mathbf{x} - \mathbf{b}\|_2 = \frac{\mathbf{x}-\mathbf{b}}{\|\mathbf{x}-\mathbf{b}\|_2}$

max. 2 A4-pages, 11pt!

5. Basics

Prod.: $\mathbb{P}(X,Y) = \mathbb{P}(Y)\,\mathbb{P}(X \mid Y) = \mathbb{P}(X)\,\mathbb{P}(Y \mid X)$

Chain: $\mathbb{P}(X_1,\dots,X_n) \equiv \mathbb{P}(X_{1:n}) = \mathbb{P}(X_1)\,\mathbb{P}(X_2 \mid X_1)\dots\mathbb{P}(X_n \mid X_{1:n-1})$

Marginal: $\mathbb{P}(X_{1:n}) = \sum_y \mathbb{P}(X_{1:n}, Y{=}y)$

$X \perp Y$: $\mathbb{P}(X \mid Y) = \mathbb{P}(X)$, $\mathbb{P}(X,Y) = \mathbb{P}(X)\,\mathbb{P}(Y)$

Bayes: $\mathbb{P}(X \mid Y) = \frac{\mathbb{P}(X,Y)}{\mathbb{P}(Y)} = \frac{\mathbb{P}(X)\,\mathbb{P}(Y|X)}{\mathbb{P}(Y)}$

Variance: $\mathbb{V}[X] = \mathbb{E}[(X-\mu_X)^2] = \mathbb{E}[X^2] - \mu_X^2$, $\mathbb{V}[X+Y] = \mathbb{V}[X] + \mathbb{V}[Y] + 2\,\mathbb{C}\mathrm{ov}(X,Y)$

C.: $\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{E}[(X-\mu_X)(Y-\mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$, $\mathbb{C}\mathrm{ov}(aX,bY) = ab\,\mathbb{C}\mathrm{ov}(X,Y)$

CoV: $Y = g(X)$, $f_Y(y) = f_X(g^{-1}(y)) \cdot \left|\frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y)\right|$

Gauss: $\mathcal{N} = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})\right)$

Condit.: $\mathbf{X}_A | X_B{=}\mathbf{x}_B \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}, \tilde{\Sigma})$ where $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu}_A + \Sigma_{AB}\Sigma_{BB}^{-1}(\mathbf{x}_B - \boldsymbol{\mu}_B)$ and $\tilde{\Sigma} = \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$

Compl. the sq.: If $p(\mathbf{x}) \propto \exp(-\frac{1}{2}\mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{x}^\top \mathbf{b})$, then $p(\mathbf{x}) = \mathcal{N}(\mathbf{x} \mid \mathbf{A}^{-1}\mathbf{b}, \mathbf{A}^{-1})$

KL-Div.: $D_{\mathrm{KL}}(p \parallel q) = \sum_x p(x) \log\left(\frac{p(x)}{q(x)}\right) \geq 0$

Entropy: $H(X) = \mathbb{E}_{x\sim p}[-\log p(x)]$;
$H(\prod_i p_i) = \sum_i H[p_i]$; $H(X,Y) = H(X) + H(Y \mid X)$;
$H(S \mid T) \geq H(S \mid T, U)$ "information never hurts";
$H(\mathcal{N}(\boldsymbol{\mu}, \Sigma)) = \frac{1}{2}\log|2\pi\,\mathrm{e}\,\Sigma|$;

Orthogonal A: $A^{-1} = A^\top$; $AA^\top = A^\top A = \|A\|_2^2 = \mathbf{I}$;
$\det(A) \in \{\pm 1\}$; $A^{-\top} = (A^\top)^{-1}$; $\mathrm{rank}(A) = n$

Inverse: $A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Deriv: $(fg)' = f'g + fg'$; $(f/g)' = (f'g - fg')/g^2$;
$f(g(x))' = f'(g(x))g'(x)$; $\log'(x) = 1/x$;
$\partial/\partial\mathbf{x}(\mathbf{b}^\top\mathbf{x}) = \mathbf{b}$; $\partial/\partial\mathbf{x}(\mathbf{x}^\top\mathbf{A}\mathbf{x}) = (\mathbf{A}^\top + \mathbf{A})\mathbf{x} \overset{*}{=} 2\mathbf{A}\mathbf{x}$
$*$: $\mathbf{A}$ sym.; $\partial/\partial\mathbf{x}(\mathbf{c}^\top\mathbf{X}\mathbf{b}) = \mathbf{c}\mathbf{b}^\top$; $\partial/\partial\mathbf{x}\,Tr(\mathbf{X}^\top\mathbf{A}) = \mathbf{A}$;
$\partial/\partial\mathbf{x}\log|\mathbf{X}| = \mathbf{X}^{-\top}$; $\partial/\partial\mathbf{x}\|\mathbf{x}-\mathbf{b}\|_2 = \frac{\mathbf{x}-\mathbf{b}}{\|\mathbf{x}-\mathbf{b}\|_2}$

Convex: $g(x)$ is conv. $\iff$ $x_1, x_2 \in \mathbb{R}$, $\lambda \in [0,1]$:
$g''(x) > 0$; $g(\lambda x_1 + (1-\lambda)x_2) \leq \lambda g(x_1) + (1-\lambda)g(x_2)$

Jensen ineqality: g convex: $g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$;
g concave (e.g. log): $g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$

Bayesian Learning: Prior $p(\theta)$;
Likelihood $p(y_{1:n} \mid x_{1:n}, \theta) \overset{iid}{=} \prod_{i=1}^n (\dots)$;
Posterior $p(\theta \mid x_{1:n}, y_{1:n}) = \frac{1}{Z} p(\theta) \prod_{i=1}^n (\dots)$;
Evidence $Z \neq 0$; **Prediction**
$p(y^* \mid x^*, x_{1:n}, y_{1:n}) = \int p(y^* \mid x^*, \theta)\, p(\theta \mid x_{1:n}, y_{1:n})\, \mathrm{d}\theta$

6. Regression